

Understanding Analysis, 2nd Edition

Chapter 4 - Section 4.5 - The Intermediate Value Theorem

1. Show how the IVT follows as a corollary to Theorem 4.5.2.

Solution: Theorem 4.5.2: continuous function preserves connected sets.

Definition of connected sets: A set $E \subseteq \mathbf{R}$ is connected iff whenever $a < c < b$ with $a, b \in E$, then $c \in E$ as well.

In $\mathbf{R}$, connected sets coincide precisely with the collection of intervals (page 104).

Then IVT follows pretty much by definition. A closed interval $[a, b]$ is connected. Let f be continuous, then $f([a, b])$ is connected. By definition of connectedness, f contains everything in $[f(a), f(b)]$.

2. Example or prove impossible.
 - (a) A continuous function defined on an open interval with range equal to a closed interval.
 - (b) A continuous function defined on a closed interval with range equal to an open interval.
 - (c) A continuous function defined on an open interval with range equal to an unbounded closed set different from $\mathbf{R}$.
 - (d) A continuous function defined on all of $\mathbf{R}$ with range equal to $\mathbf{Q}$.

Solution:

(a) $\sin(1/x)$ is continuous on $(0, 1)$ with range $[-1, 1]$.

(b) For unbounded interval, yes. $f(x) = x$ is continuous on $\mathbf{R}$ and its range is also $\mathbf{R}$ (recall that $\mathbf{R}$ is both open and closed).

For unbounded interval, impossible. A closed bounded interval is compact and continuous functions preserve compactness (Theorem 4.4.1).

(c) With unbounded interval: x^2 is continuous on $\mathbf{R}$ with range $[0, \infty)$.

With bounded interval: $\tan^2 x$ is continuous on $(\frac{\pi}{2}, \frac{\pi}{2})$ with range $[0, \infty)$.

(d) Impossible. We'll prove it by contradiction. Suppose there exists such a function. Since the range is $\mathbf{Q}$, f is not constant. Choose $x \neq y$ such that $f(x) \neq f(y)$ and wlog $f(x) < f(y)$. By IVT, the range of f must contain everything in $[f(x), f(y)]$. But this implies f contains an irrational, since between any two rationals there is an irrational, contradicting the assumption that the range is $\mathbf{Q}$.

3. A function f is *increasing* on A if $f(x) \leq f(y)$ for all $x < y$ in A . Show that if f is increasing on $[a, b]$ and satisfies the intermediate value property (Definition 4.5.3), then f is continuous on $[a, b]$.

Solution: To prove that f is continuous at $a \leq c \leq b$, we need to show that given $\epsilon > 0$, there exists $\delta > 0$ such that $|x - c| < \delta \implies |f(x) - f(c)| < \epsilon$.

We'll prove left continuity: there exists δ_a such that for $x < c$ and $c - x < \delta_a$, $f(c) - f(x) < \epsilon$. The proof for $x > c$ is similar and will result in a δ_b . Then choose $\delta = \min(\delta_a, \delta_b)$.

By monotonicity, $f(a) \leq f(c)$ and if $a < x < c$ then $f(a) \leq f(x) \leq f(c)$. By intermediate value property, f takes on all values between $f(a)$ and $f(c)$ so there is an $a < x < c$ such that $f(c) - f(x) < \epsilon$. Choose $\delta_a = c - x$. This works for all $x < t < c$ as well because monotonicity ensures that $f(c) - f(t) \leq f(c) - f(x) < \epsilon$.

4. Let g be continuous on an interval A and let F be the set of points where g fails to be one-to-one; that is,

$$F = \{x \in A : f(x) = f(y) \text{ for some } y \neq x \text{ and } y \in A\}.$$

Show F is either empty or uncountable.

Solution: At first I thought the problem statement must be wrong. Take something like $\sin(x)$. Within an interval there's only a finite number of points where $\sin(x) = 1$, so it's neither empty nor uncountable. But then I realized it's not only 1 that's repeating, the intervals around $\sin(x) = 1$ must be repeating as well and an interval is uncountable. That's what we'll prove; F is either empty or contains repeating intervals of output values.

If g never fails to be one-to-one then F is trivially empty. Otherwise there exists $x \neq y$ such that $g(x) = g(y)$.

If for all $x < c < y$, $g(c) = g(x) = g(y)$, then $[x, y] \subseteq F$ so F is uncountable.

Otherwise, let $g(c) \neq g(x)$ and wlog $g(c) > g(x)$. By IVT there exists a subset $S_1 \subseteq (x, c)$ such that $g(S_1) = (g(x), g(c))$. Since $g(S_1)$ is an interval, S_1 must be uncountable.

Similarly, there exists a subset $S_2 \subseteq (c, y)$ such that $g(S_2) = (g(y), g(c))$.

Since $g(x) = g(y)$ then $g(S_1) = g(S_2)$, hence $S_1, S_2 \subseteq F$ and F is uncountable. Note that $S_1 \cap S_2 = \emptyset$ so they're different set of points and can be included in F .

5. (a) Finish the proof of the Intermediate Value Theorem using the Axiom of Completeness started previously.
- (b) Finish the proof of the Intermediate Value Theorem using the Nested Interval Property started previously.

Solution:

- (a) The partial proof from the book is as follow. Let f be continuous and consider the special case $f(a) < 0 < f(b)$. We'll show that $f(c) = 0$ for some $c \in (a, b)$.
Let

$$K = \{x \in [a, b] : f(x) \leq 0\}.$$

Notice that K is bounded above by b and $a \in K$ so K is not empty. Thus we may appeal to AoC to assert that $s = \sup K$ exists.

We'll first prove that $f(s) \leq 0$. Since s is a supremum, s is a limit point of some sequence $(x_n) \in K$. By continuity, $f(x_n) \rightarrow f(s)$. Since $f(x_n) \leq 0$, then $f(s) \leq 0$.

Next we need to prove that $f(s) \geq 0$. The idea is that since s is a supremum, it's sort of the "rightmost" point where f is ≤ 0 . But if $f(s) < 0$, since everything to the right of s is > 0 , there'd be a discontinuity; a jump that skips 0. Or in other words, $f(s)$ is not continuous from the right side.

Formally, we prove by contradiction. First we note that since $s = \sup K$, if $x > s$ then $x \notin K$ hence $f(x) > 0$.

Now let $\epsilon = |f(s)|$. By continuity, there exists a $\delta < (b - s)$ that works (the bound $(b - s)$ is just a trivial detail to ensure we stay within $[a, b]$). Choose x such that $x - s < \delta$. Since f is defined on $[a, b]$, $f(x)$ is defined.

We've asserted that $f(x) > 0$. But since $f(s) < 0$, we have $f(x) - f(s) \geq |0 - f(s)| = \epsilon$, contradicting continuity of f .

- (b) Repeat the procedure described in the book to get a sequence of nested intervals. Apply NIP. It's not difficult to conclude that the intersection is 0.

6. Let $f : [0, 1] \rightarrow \mathbf{R}$ be continuous with $f(0) = f(1)$.

- (a) Show that there must exist $x, y \in [0, 1]$ satisfying $|x - y| = 1/2$ and $f(x) = f(y)$.
 (b) Show that for each $n \in \mathbf{N}$ there exist $x_n, y_n \in [0, 1]$ with $|x_n - y_n| = 1/n$ and $f(x_n) = f(y_n)$.
 (c) If $h \in (0, 1/2)$ is not of the form $1/n$, there does not necessarily exist $|x - y| = h$ satisfying $f(x) = f(y)$. Provide an example that illustrates this using $h = 2/5$.

Solution:

- (a) Use the same trick as problem 7 (which I had solved before this). Define $g(x) = f(x) - f(x + 1/2)$ for $x \in [0, 1/2]$. Then the required condition is equivalent to the existence of $g(x) = 0$.

Consider $g(0)$ and $g(1/2)$. Since $f(0) = f(1)$, then $g(0) = -g(1/2)$ (both could possibly be 0 as well). Apply IVT to $[0, 1/2]$. Since $g(0)$ and $g(1/2)$ have different signs, there's a $g(x) = 0$.

- (b) Similar to part (a), consider the function $g(x) = f(x) - f(x + \frac{1}{n})$. Then the required condition is equivalent to the existence of $g(x) = 0$.

Assume $n = 3$ and consider $g(0), g(\frac{1}{3})$, and $g(\frac{2}{3})$. If any of them are zero then we're done. Otherwise, like part (a) we'll prove that there exist $i < j$ such that $g(i)$ and $g(j)$ have different signs. Then we'll apply IVT to conclude the existence of $g(x) = 0$ for some x .

Is it possible for all three g 's to be positive? Suppose $g(0)$ and $g(\frac{1}{3})$ are positive, then we can conclude that $f(0) > f(\frac{1}{3}) > f(\frac{2}{3})$. But $f(0) = f(1)$, so we must have $f(1) > f(\frac{2}{3}) \implies g(\frac{2}{3})$ is negative.

A very similar argument can be used to prove that we can't have all three g 's negative, and it's also easy to generalize to arbitrary n .

Equivalently, we can express the same argument with telescoping sum. Consider

$$\sum_{k=0}^{n-1} g\left(\frac{k}{n}\right) = \sum_{k=0}^{n-1} \left[f\left(\frac{k}{n}\right) - f\left(\frac{k+1}{n}\right) \right]$$

This sum telescopes to $f(0) - f(1) = 0$. Since the sum is zero, we can't have all terms positive or all terms negative.

(c) TODO.

7. Let f be a continuous function on the closed interval $[0, 1]$ with range also contained in $[0, 1]$. Prove that f must have a fixed point; that is, show $f(x) = x$ for at least one value of $x \in [0, 1]$.

Solution: Stuck with this for quite some time and finally got a hint from chatgpt: consider the auxiliary function $g(x) = f(x) - x$.

With that hint, the problem becomes much easier. $f(x) = x$ if $g(x) = 0$. Since the range of f is in $[0, 1]$, then $0 \leq f(0), f(1) \leq 1$ and $g(1) \leq 0 \leq g(0)$. Note that g is continuous, so we can apply IVT to $[0, 1]$ and conclude there exists an $0 \leq x \leq 1$ such that $g(x) = 0$.

8. **(Inverse functions).** If a function $f : A \rightarrow R$ is one-to-one (injective), then we can define the inverse function f^{-1} on the range of f in the natural way: $f^{-1}(y) = x$ where $y = f(x)$. Show that if f is continuous on an interval $[a, b]$ and one-to-one, then f^{-1} is also continuous.

Solution: Take a convergent sequence $(y_n) \rightarrow y$ and let $f^{-1}(y_n) = (x_n)$. To prove continuity of f^{-1} , we need to prove that $(y_n) \rightarrow y \implies (x_n) \rightarrow x$.

Wrong solution

My first stupid thought was to argue that since f is continuous, then $(x_n) \rightarrow x \implies (f(x_n) = (y_n)) \rightarrow (f(x) = y)$. But this argument is circular; we're using the thing we're trying to prove, mainly that $(x_n) \rightarrow x$. We *don't know* that $(x_n) \rightarrow x$.

Correct solution

In fact, we don't know yet that (x_n) converges at all. But we can use compactness: since $(x_n) \in [a, b]$ then there's a convergent subsequence $(x_{n_k}) \rightarrow L$.

Our strategy will be to prove that $L = x$. Then we show that the whole sequence $(x_n) \rightarrow x$.

By continuity of f , we have $(x_{n_k}) \rightarrow L \implies f(x_{n_k}) \rightarrow f(L)$. But note that since $f(x_{n_k})$ is really a subsequence of (y_n) and by assumption $(y_n) \rightarrow y$, then $f(x_{n_k}) \rightarrow f(L) = y = f(x)$, hence $L = x$.

Note that the above argument works for any arbitrary convergent subsequence $(x_{n_k}) \in (x_n)$. Finally, we invoke Exercise 2.5.5 (which I haven't solved!): if a bounded sequence (x_n) has the property that every convergent subsequence converges to the same limit $L \in \mathbf{R}$, then $(x_n) \rightarrow L$. Here (x_n) is bounded by $[a, b]$ and $L = x$.